Lecture-1. Python fundamentals.

Python is a high-level, interpreted programming language known for its simplicity and readability. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Python is widely used in various fields such as web development, data science, artificial intelligence, and more.

Python Intro

Python was created by Guido van Rossum and first released in 1991. It emphasizes readability and simplicity, allowing developers to focus more on problem-solving than complex syntax. Python is known for its large standard library and vast ecosystem of third-party packages.

Why Python?

- Easy to read and learn
- Strong community support with many libraries and frameworks
- Works on all major platforms (Windows, macOS, Linux)

Python User Input

In Python, user input can be captured using the `input()` function. This function waits for the user to type something and press Enter. The input is always returned **as a string**.

- **Syntax:**

```
name = input("Enter your name: ")
print("Hello", name)
```

- To work with numerical inputs, you need to cast the input to the appropriate type.

Example:

```
age = int(input("Enter your age: ")) # Converts string input to integer
print("You are " + str(age) + " years old.") # Print eith integer
```

Python Get Started

To get started with Python, you need to install the Python interpreter, which is available at python.org. Once installed, you can run Python programs in an Integrated Development Environment (IDE) like PyCharm, VSCode, or Jupyter notebooks, or directly through the terminal.

Python Syntax

Python syntax refers to the set of rules that define how Python programs are written. For example:

- Statements are usually written on a single line.
- Indentation is crucial as it defines code blocks (e.g., loops, conditionals, functions).
- Comments start with `#` for single-line comments.

Python Comments

Comments are used to explain code. Python supports single-line comments using the `#` symbol. Multi-line comments can be created using triple quotes (`'''` or `"""`).

```
# This is a single-line comment
'''
This is a
multi-line comment
'''
```

Python Variables

A variable in Python is used to store data that can be referenced later. Python is dynamically typed, meaning you don't need to declare a variable's type.

```
x = 5
x = "KBTU"
print(x)

name = "Alice"
```

Python Data Types

Python supports various data types, including:

- `int` (integer numbers)
- `float` (floating-point numbers)
- `str` (strings)
- `bool` (boolean values)
- `list` (ordered collections)
- `tuple` (immutable ordered collections)
- `dict` (key-value pairs)

Python Numbers

Python supports both integers (`int`) and floating-point numbers (`float`). Arithmetic operations like addition, subtraction, multiplication, and division can be performed on numbers.

```
x = 10
y = 2.5
sum = x + y  # 12.5
```

Python Casting

Casting allows you to convert between different data types. This is done using the `int()`, `float()`, or `str()` functions.

```
x = 10.5
y = int(x)  # 10
```

Python Strings

Strings are sequences of characters enclosed in single, double, or triple quotes. You can manipulate strings using various string methods.

```
text = "Hello, World!"
print(text[0])  # H
```

Python String Formatting

Python provides several ways to format strings:

1. Using the `%` operator
2. Using the `format()` method
3. Using f-strings (available in Python 3.6+)

```
name = "Alice"
age = 25
print(f"Hello, {name}. You are {age} years old.")
```

Python Booleans

Booleans represent truth values. The two boolean values are `True` and `False`. They are often used in conditional statements and comparisons.

```
is_adult = True
is_student = False
```

Python Operators

Python supports a variety of operators:

- Arithmetic: `+`, ` `, ` `, `/`, `%`, `//`, `**`
- Comparison: `==`, `!=`, `>`, `<`, `>=`, `<=`
- Logical: `and`, `or`, `not`
- Assignment: `=`, `+=`, -`=`, *`=`, `/=`

Python If...Else

The `if` statement allows you to execute a block of code if a condition is `True`. The `else` block runs if the condition is `False`.

```
age = 20
if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")
```

Git

Git is a version control system that tracks changes in source code during software development. It allows developers to collaborate efficiently, manage different versions of a project, and roll back changes if necessary.

- Key commands:
    - `git init`: Initializes a new Git repository.
    - `git clone <repo_url>`: Clones an existing repository.
    - `git add .`: Stages changes for commit.
    - `git commit -m "message"`: Commits staged changes.
    - `git push`: Pushes commits to the remote repository.
    - `git pull`: Fetches updates from the remote repository.